

Series : ZXW4Y



SET~1

प्रश्न-पत्र कोड
Q.P. Code

65/4/1

रोल नं.

Roll No.



परीक्षार्थी प्रश्न-पत्र कोड को उत्तर-पुस्तिका के मुख-पृष्ठ पर अवश्य लिखें।

Candidates must write the Q.P. Code on the title page of the answer-book.



गणित

MATHEMATICS



निर्धारित समय : 3 घण्टे

Time allowed : 3 hours

अधिकतम अंक : 80

Maximum Marks : 80

नोट

- (I) कृपया जाँच कर लें कि इस प्रश्न-पत्र में मुद्रित पृष्ठ 23 हैं।
- (II) प्रश्न-पत्र में दाहिने हाथ की ओर दिए गए प्रश्न-पत्र कोड को परीक्षार्थी उत्तर-पुस्तिका के मुख-पृष्ठ पर लिखें।
- (III) कृपया जाँच कर लें कि इस प्रश्न-पत्र में 38 प्रश्न हैं।
- (IV) कृपया प्रश्न का उत्तर लिखना शुरू करने से पहले, उत्तर-पुस्तिका में यथा स्थान पर प्रश्न का क्रमांक अवश्य लिखें।
- (V) इस प्रश्न-पत्र को पढ़ने के लिए 15 मिनट का समय दिया गया है। प्रश्न-पत्र का वितरण पूर्वाह्न में 10.15 बजे किया जाएगा। 10.15 बजे से 10.30 बजे तक परीक्षार्थी केवल प्रश्न-पत्र को पढ़ेंगे और इस अवधि के दौरान वे उत्तर-पुस्तिका पर कोई उत्तर नहीं लिखेंगे।

NOTE

- (I) Please check that this question paper contains 23 printed pages.
- (II) Q.P. Code given on the right hand side of the question paper should be written on the title page of the answer-book by the candidate.
- (III) Please check that this question paper contains 38 questions.
- (IV) Please write down the Serial Number of the question in the answer-book at the given place before attempting it.
- (V) 15 minute time has been allotted to read this question paper. The question paper will be distributed at 10.15 a.m. From 10.15 a.m. to 10.30 a.m., the candidates will read the question paper only and will not write any answer on the answer-book during this period. #



सामान्य निर्देश :

निम्नलिखित निर्देशों को बहुत सावधानी से पढ़िए और उनका सख्ती से पालन कीजिए :

- इस प्रश्न-पत्र में 38 प्रश्न हैं। सभी प्रश्न अनिवार्य हैं।
- यह प्रश्न-पत्र पाँच खण्डों में विभाजित है – क, ख ग, घ एवं ङ।
- खण्ड क में प्रश्न संख्या 1 से 18 तक बहुविकल्पीय (MCQ) तथा प्रश्न संख्या 19 एवं 20 अभिकथन एवं तर्क आधारित 1 अंक के प्रश्न हैं।
- खण्ड ख में प्रश्न संख्या 21 से 25 तक अति लघु-उत्तरीय (VSA) प्रकार के 2 अंकों के प्रश्न हैं।
- खण्ड ग में प्रश्न संख्या 26 से 31 तक लघु-उत्तरीय (SA) प्रकार के 3 अंकों के प्रश्न हैं।
- खण्ड घ में प्रश्न संख्या 32 से 35 तक दीर्घ-उत्तरीय (LA) प्रकार के 5 अंकों के प्रश्न हैं।
- खण्ड ङ में प्रश्न संख्या 36 से 38 तक प्रकरण अध्ययन आधारित 4 अंकों के प्रश्न हैं।
- प्रश्न-पत्र में समग्र विकल्प नहीं दिया गया है। यद्यपि, खण्ड ख के 2 प्रश्नों में, खण्ड ग के 3 प्रश्नों में, खण्ड घ के 2 प्रश्नों में तथा खण्ड ङ के 2 प्रश्नों में आंतरिक विकल्प का प्रावधान दिया गया है।
- कैल्कुलेटर का उपयोग वर्जित है।

खण्ड क

इस खण्ड में बहुविकल्पीय प्रश्न (MCQ) हैं, जिनमें प्रत्येक प्रश्न 1 अंक का है।

1. $\sin^{-1}\left(\sin\left(-\frac{10\pi}{3}\right)\right)$ का मुख्य मान है :

(A) $-\frac{2\pi}{3}$

(B) $-\frac{\pi}{3}$

(C) $\frac{\pi}{3}$

(D) $\frac{2\pi}{3}$

2. यदि A तथा B समान कोटि के वर्ग आव्यूह हैं तथा $AB = A$ तथा $BA = B$ है, तो $A^2 + B^2$ बराबर है :

(A) $A + B$

(B) BA

(C) $2(A + B)$

(D) $2BA$

3. माना वास्तविक x के लिए, $f(x) = x^3 + 5x + 1$ है, तो :

(A) f, R पर एकैकी है लेकिन आच्छादक नहीं है

(B) f, R पर आच्छादक है लेकिन एकैकी नहीं है

(C) f, R पर एकैकी और आच्छादक है

(D) f, R पर न तो एकैकी है और न ही आच्छादक



General Instructions :

Read the following instructions very carefully and strictly follow them :

- (i) This question paper contains **38** questions. **All** questions are **compulsory**.
- (ii) This question paper is divided into **five** Sections – **A, B, C, D** and **E**.
- (iii) In **Section A**, Questions no. **1** to **18** are multiple choice questions (MCQs) and questions number **19** and **20** are Assertion-Reason based questions of **1** mark each.
- (iv) In **Section B**, Questions no. **21** to **25** are very short answer (VSA) type questions, carrying **2** marks each.
- (v) In **Section C**, Questions no. **26** to **31** are short answer (SA) type questions, carrying **3** marks each.
- (vi) In **Section D**, Questions no. **32** to **35** are long answer (LA) type questions carrying **5** marks each.
- (vii) In **Section E**, Questions no. **36** to **38** are case study based questions carrying **4** marks each.
- (viii) There is no overall choice. However, an internal choice has been provided in 2 questions in Section B, 3 questions in Section C, 2 questions in Section D and 2 questions in Section E.
- (ix) Use of calculator is **not** allowed.

SECTION A

This section comprises multiple choice questions (MCQs) of 1 mark each.

1. The principal value of $\sin^{-1}\left(\sin\left(-\frac{10\pi}{3}\right)\right)$ is :
(A) $-\frac{2\pi}{3}$ (B) $-\frac{\pi}{3}$
(C) $\frac{\pi}{3}$ (D) $\frac{2\pi}{3}$
2. If A and B are square matrices of same order such that $AB = A$ and $BA = B$, then $A^2 + B^2$ is equal to :
(A) $A + B$ (B) BA
(C) $2(A + B)$ (D) $2BA$
3. For real x, let $f(x) = x^3 + 5x + 1$. Then :
(A) f is one-one but not onto on R
(B) f is onto on R but not one-one
(C) f is one-one and onto on R
(D) f is neither one-one nor onto on R



4. यदि $y = \sin^{-1} x$ है, तो $(1 - x^2) \frac{d^2 y}{dx^2}$ बराबर है :
- (A) $x \frac{dy}{dx}$ (B) $-x \frac{dy}{dx}$
- (C) $x^2 \frac{dy}{dx}$ (D) $-x^2 \frac{dy}{dx}$
5. λ के वे मान जिनके लिए $f(x) = \sin x - \cos x - \lambda x + C$, x के सभी वास्तविक मानों के लिए हासमान है, हैं :
- (A) $1 < \lambda < \sqrt{2}$ (B) $\lambda \geq 1$
- (C) $\lambda \geq \sqrt{2}$ (D) $\lambda < 1$
6. बिंदुओं $(3, 6, -1)$ तथा $(6, 2, -2)$ को मिलाने वाले रेखाखण्ड पर एक बिंदु P ऐसा है कि इसका y -निर्देशांक 4 है, तो इसका z -निर्देशांक है :
- (A) $-\frac{3}{2}$ (B) 0
- (C) 1 (D) $\frac{3}{2}$
7. यदि M तथा N कोटि 3 के ऐसे वर्ग आव्यूह हैं कि $\det(M) = m$ तथा $MN = mI$ है, तो $\det(N)$ बराबर है :
- (A) -1 (B) 1
- (C) $-m^2$ (D) m^2
8. यदि $f(x) = \begin{cases} 3x - 2, & 0 < x \leq 1 \\ 2x^2 + ax, & 1 < x < 2 \end{cases}$ $x \in (0, 2)$ में संतत है, तो a बराबर है :
- (A) -4 (B) $-\frac{7}{2}$
- (C) -2 (D) -1



4. If $y = \sin^{-1} x$, then $(1 - x^2) \frac{d^2 y}{dx^2}$ is equal to :
- (A) $x \frac{dy}{dx}$ (B) $-x \frac{dy}{dx}$
- (C) $x^2 \frac{dy}{dx}$ (D) $-x^2 \frac{dy}{dx}$
5. The values of λ so that $f(x) = \sin x - \cos x - \lambda x + C$ decreases for all real values of x are :
- (A) $1 < \lambda < \sqrt{2}$ (B) $\lambda \geq 1$
- (C) $\lambda \geq \sqrt{2}$ (D) $\lambda < 1$
6. If P is a point on the line segment joining $(3, 6, -1)$ and $(6, 2, -2)$ and y-coordinate of P is 4, then its z-coordinate is :
- (A) $-\frac{3}{2}$ (B) 0
- (C) 1 (D) $\frac{3}{2}$
7. If M and N are square matrices of order 3 such that $\det(M) = m$ and $MN = mI$, then $\det(N)$ is equal to :
- (A) -1 (B) 1
- (C) $-m^2$ (D) m^2
8. If $f(x) = \begin{cases} 3x - 2, & 0 < x \leq 1 \\ 2x^2 + ax, & 1 < x < 2 \end{cases}$ is continuous for $x \in (0, 2)$, then a is equal to :
- (A) -4 (B) $-\frac{7}{2}$
- (C) -2 (D) -1



9. यदि $f: \mathbb{N} \rightarrow \mathbb{W}$,

$$f(n) = \begin{cases} \frac{n}{2}, & \text{यदि } n \text{ सम है} \\ 0, & \text{यदि } n \text{ विषम है} \end{cases}$$

द्वारा परिभाषित है, तो f :

- (A) केवल एकैकी है (B) केवल आच्छादी है
(C) एकैकी-आच्छादी है (D) न तो आच्छादी है और न ही एकैकी

10. आव्यूह $\begin{bmatrix} 0 & 1 & -2 \\ -1 & 0 & -7 \\ 2 & 7 & 0 \end{bmatrix}$ है एक :

- (A) विकर्ण आव्यूह (B) सममित आव्यूह
(C) विषम सममित आव्यूह (D) अदिश आव्यूह

11. यदि ΔABC की भुजाएँ AB तथा AC क्रमशः सदिशों $\hat{j} + \hat{k}$ तथा $3\hat{i} - \hat{j} + 4\hat{k}$ द्वारा निरूपित हैं, तो A से BC पर खींची गई माध्यिका की लंबाई है :

- (A) $2\sqrt{2}$ इकाई (B) $\sqrt{18}$ इकाई
(C) $\frac{\sqrt{34}}{2}$ इकाई (D) $\frac{\sqrt{48}}{2}$ इकाई

12. $f(x) = \begin{cases} x, & \text{यदि } x \leq 1 \\ 5, & \text{यदि } x > 1 \end{cases}$

द्वारा परिभाषित फलन f संतत नहीं है :

- (A) $x = 0$ पर (B) $x = 1$ पर
(C) $x = 2$ पर (D) $x = 5$ पर

13. यदि $f(x) = 2x + \cos x$ है, तो $f(x)$:

- (A) का $x = \pi$ पर अधिकतम मान है (B) का $x = \pi$ पर न्यूनतम मान है
(C) एक वर्धमान फलन है (D) एक हासमान फलन है



9. If $f : N \rightarrow W$ is defined as

$$f(n) = \begin{cases} \frac{n}{2}, & \text{if } n \text{ is even} \\ 0, & \text{if } n \text{ is odd} \end{cases},$$

then f is :

- (A) injective only (B) surjective only
(C) a bijection (D) neither surjective nor injective

10. The matrix $\begin{bmatrix} 0 & 1 & -2 \\ -1 & 0 & -7 \\ 2 & 7 & 0 \end{bmatrix}$ is a :

- (A) diagonal matrix (B) symmetric matrix
(C) skew symmetric matrix (D) scalar matrix

11. If the sides AB and AC of ΔABC are represented by vectors $\hat{j} + \hat{k}$ and $3\hat{i} - \hat{j} + 4\hat{k}$ respectively, then the length of the median through A on BC is :

- (A) $2\sqrt{2}$ units (B) $\sqrt{18}$ units
(C) $\frac{\sqrt{34}}{2}$ units (D) $\frac{\sqrt{48}}{2}$ units

12. The function f defined by

$$f(x) = \begin{cases} x, & \text{if } x \leq 1 \\ 5, & \text{if } x > 1 \end{cases}$$

is **not** continuous at :

- (A) $x = 0$ (B) $x = 1$
(C) $x = 2$ (D) $x = 5$

13. If $f(x) = 2x + \cos x$, then $f(x)$:

- (A) has a maxima at $x = \pi$ (B) has a minima at $x = \pi$
(C) is an increasing function (D) is a decreasing function



14. $\int \frac{\cos 2x - \cos 2\alpha}{\cos x - \cos \alpha} dx$ बराबर है :

- (A) $2(\sin x + x \cos \alpha) + C$ (B) $2(\sin x - x \cos \alpha) + C$
 (C) $2(\sin x + 2x \cos \alpha) + C$ (D) $2(\sin x + \sin \alpha) + C$

15. $\int_0^1 \frac{dx}{e^x + e^{-x}}$ का मान है :

- (A) $-\frac{\pi}{4}$ (B) $\frac{\pi}{4}$
 (C) $\tan^{-1} e - \frac{\pi}{4}$ (D) $\tan^{-1} e$

16. अवकल समीकरण $\left(\frac{d^2y}{dx^2}\right)^2 + \left(\frac{dy}{dx}\right)^2 = x \sin\left(\frac{dy}{dx}\right)$ की कोटि तथा घात है :

- (A) कोटि 2, घात 2 (B) कोटि 2, घात 1
 (C) कोटि 2, घात परिभाषित नहीं (D) कोटि 1, घात परिभाषित नहीं

17. वक्र $y = \sqrt{x}$, रेखाओं $x = 0$ तथा $x = 4$ तथा x -अक्ष से घिरे क्षेत्र का क्षेत्रफल है :

- (A) $\frac{16}{9}$ वर्ग इकाई (B) $\frac{32}{9}$ वर्ग इकाई
 (C) $\frac{16}{3}$ वर्ग इकाई (D) $\frac{32}{3}$ वर्ग इकाई

18. एक रैखिक प्रोग्रामन समस्या (LPP) के सुसंगत क्षेत्र के कोनीय बिंदु $(0, 2)$, $(3, 0)$, $(6, 0)$, $(6, 8)$ तथा $(0, 5)$ हैं। यदि उद्देश्य फलन $Z = ax + by$; $(a, b > 0)$ का अधिकतम मान $(0, 2)$ तथा $(3, 0)$ पर प्राप्त होता है, तो a तथा b में संबंध है :

- (A) $a = b$ (B) $a = 3b$
 (C) $b = 6a$ (D) $3a = 2b$



14. $\int \frac{\cos 2x - \cos 2\alpha}{\cos x - \cos \alpha} dx$ is equal to :

- (A) $2(\sin x + x \cos \alpha) + C$ (B) $2(\sin x - x \cos \alpha) + C$
(C) $2(\sin x + 2x \cos \alpha) + C$ (D) $2(\sin x + \sin \alpha) + C$

15. The value of $\int_0^1 \frac{dx}{e^x + e^{-x}}$ is :

- (A) $-\frac{\pi}{4}$ (B) $\frac{\pi}{4}$
(C) $\tan^{-1} e - \frac{\pi}{4}$ (D) $\tan^{-1} e$

16. The order and degree of the differential equation

$$\left(\frac{d^2y}{dx^2}\right)^2 + \left(\frac{dy}{dx}\right)^2 = x \sin\left(\frac{dy}{dx}\right) \text{ are :}$$

- (A) order 2, degree 2 (B) order 2, degree 1
(C) order 2, degree not defined (D) order 1, degree not defined

17. The area of the region enclosed by the curve $y = \sqrt{x}$ and the lines $x = 0$ and $x = 4$ and x-axis is :

- (A) $\frac{16}{9}$ sq. units (B) $\frac{32}{9}$ sq. units
(C) $\frac{16}{3}$ sq. units (D) $\frac{32}{3}$ sq. units

18. The corner points of the feasible region of a Linear Programming Problem are (0, 2), (3, 0), (6, 0), (6, 8) and (0, 5). If $Z = ax + by$; ($a, b > 0$) be the objective function, and maximum value of Z is obtained at (0, 2) and (3, 0), then the relation between a and b is :

- (A) $a = b$ (B) $a = 3b$
(C) $b = 6a$ (D) $3a = 2b$



प्रश्न संख्या 19 और 20 अभिकथन एवं तर्क आधारित प्रश्न हैं। दो कथन दिए गए हैं जिनमें एक को अभिकथन (A) तथा दूसरे को तर्क (R) द्वारा अंकित किया गया है। इन प्रश्नों के सही उत्तर नीचे दिए गए कोडों (A), (B), (C) और (D) में से चुनकर दीजिए।

- (A) अभिकथन (A) और तर्क (R) दोनों सही हैं और तर्क (R), अभिकथन (A) की सही व्याख्या करता है।
- (B) अभिकथन (A) और तर्क (R) दोनों सही हैं, परन्तु तर्क (R), अभिकथन (A) की सही व्याख्या नहीं करता है।
- (C) अभिकथन (A) सही है, परन्तु तर्क (R) गलत है।
- (D) अभिकथन (A) गलत है, परन्तु तर्क (R) सही है।

19. अभिकथन (A) : यदि A तथा B दो ऐसी घटनाएँ हैं कि $P(A \cap B) = 0$ है, तो A तथा B स्वतंत्र घटनाएँ हैं।
- तर्क (R) : दो घटनाएँ स्वतंत्र होती हैं यदि एक के घटने का दूसरे के घटने पर कोई प्रभाव न पड़े।
20. अभिकथन (A) : एक रैखिक प्रोग्रामन समस्या (LPP) में यदि सुसंगत क्षेत्र खाली है, तो रैखिक प्रोग्रामन समस्या (LPP) का कोई हल नहीं है।
- तर्क (R) : सुसंगत क्षेत्र वह क्षेत्र है जिसमें सारे व्यवरोध संतुष्ट होते हैं।

खण्ड ख

इस खण्ड में अति लघु-उत्तरीय (VSA) प्रकार के 5 प्रश्न हैं, जिनमें प्रत्येक के 2 अंक हैं।

21. माना A तथा B कोटि 3 के दो वर्ग आव्यूह हैं तथा $\det(A) = 3$ तथा $\det(B) = -4$ है। $\det(-6AB)$ का मान ज्ञात कीजिए।
22. (क) 'a' का वह न्यूनतम मान ज्ञात कीजिए जिसके लिए $f(x) = 2x^2 - ax + 3$; $[2, 4]$ पर एक वर्धमान फलन हो।

अथवा

- (ख) यदि $f(x) = x + \frac{1}{x}$, $x \geq 1$ है, तो दर्शाइए कि f एक वर्धमान फलन है।

23. (क) $\sin^{-1}\left(\frac{x}{\sqrt{1+x^2}}\right)$ को सरलतम रूप में लिखिए।

अथवा

- (ख) $\sin^{-1}\sqrt{x-1}$ का प्रांत ज्ञात कीजिए।



Questions number **19** and **20** are Assertion and Reason based questions. Two statements are given, one labelled Assertion (A) and the other labelled Reason (R). Select the correct answer from the codes (A), (B), (C) and (D) as given below.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).
- (B) Both Assertion (A) and Reason (R) are true, but Reason (R) is **not** the correct explanation of the Assertion (A).
- (C) Assertion (A) is true, but Reason (R) is false.
- (D) Assertion (A) is false, but Reason (R) is true.

19. Assertion (A) : If A and B are two events such that $P(A \cap B) = 0$, then A and B are independent events.

Reason (R) : Two events are independent if the occurrence of one does not effect the occurrence of the other.

20. Assertion (A) : In a Linear Programming Problem, if the feasible region is empty, then the Linear Programming Problem has no solution.

Reason (R) : A feasible region is defined as the region that satisfies all the constraints.

SECTION B

This section comprises 5 Very Short Answer (VSA) type questions of 2 marks each.

21. Let A and B be two square matrices of order 3 such that $\det(A) = 3$ and $\det(B) = -4$. Find the value of $\det(-6AB)$.

22. (a) Find the least value of 'a' so that $f(x) = 2x^2 - ax + 3$ is an increasing function on $[2, 4]$.

OR

(b) If $f(x) = x + \frac{1}{x}$, $x \geq 1$, show that f is an increasing function.

23. (a) Simplify $\sin^{-1} \left(\frac{x}{\sqrt{1+x^2}} \right)$.

OR

(b) Find domain of $\sin^{-1} \sqrt{x-1}$.



24. समाकलन के प्रयोग से, वक्र $\frac{x^2}{9} + \frac{y^2}{4} = 1$ तथा x -अक्ष के बीच घिरे क्षेत्र का क्षेत्रफल परिकलित कीजिए।
25. वक्र $y = 5x - 2x^3$ के लिए यदि x , 2 इकाई/सेकण्ड की दर से बढ़ रहा है, तो $x = 2$ पर वक्र की प्रवणता (ढाल) के बदलने की दर ज्ञात कीजिए।

खण्ड ग

इस खण्ड में लघु-उत्तरीय (SA) प्रकार के 6 प्रश्न हैं, जिनमें प्रत्येक के 3 अंक हैं।

26. (क) यदि $f : \mathbb{R}^+ \rightarrow \mathbb{R}$, $f(x) = \log_a x$ ($a > 0$ तथा $a \neq 1$) द्वारा परिभाषित है, तो सिद्ध कीजिए कि f एक एकैकी-आच्छादक फलन है।
($\mathbb{R}^+$ सभी धनात्मक वास्तविक संख्याओं का समुच्चय है।)

अथवा

- (ख) माना $A = \{1, 2, 3\}$ तथा $B = \{4, 5, 6\}$. A से B में एक संबंध R इस प्रकार परिभाषित है कि $R = \{(x, y) : x + y = 6, x \in A, y \in B\}$ है।

- (i) R के सभी अवयव लिखिए।
(ii) क्या R एक फलन है? सत्यापित कीजिए।
(iii) R का प्रांत व परिसर ज्ञात कीजिए।

27. (क) यदि फलन $f(x) = \begin{cases} \frac{x^2 - 2x - 3}{x + 1}, & x \neq -1 \\ k, & x = -1 \end{cases}$,

$x = -1$ पर संतत है, तो k का मान ज्ञात कीजिए।

अथवा

- (ख) फलन $f(x) = x |x|$ की $x = 0$ पर अवकलनीयता की जाँच कीजिए।



24. Calculate the area of the region bounded by the curve $\frac{x^2}{9} + \frac{y^2}{4} = 1$ and the x-axis using integration.
25. For the curve $y = 5x - 2x^3$, if x increases at the rate of 2 units/s, then how fast is the slope of the curve changing when $x = 2$?

SECTION C

This section comprises 6 Short Answer (SA) type questions of 3 marks each.

26. (a) If $f: \mathbb{R}^+ \rightarrow \mathbb{R}$ is defined as $f(x) = \log_a x$ ($a > 0$ and $a \neq 1$), prove that f is a bijection.
($\mathbb{R}^+$ is a set of all positive real numbers.)

OR

- (b) Let $A = \{1, 2, 3\}$ and $B = \{4, 5, 6\}$. A relation R from A to B is defined as $R = \{(x, y) : x + y = 6, x \in A, y \in B\}$.
- (i) Write all elements of R .
 - (ii) Is R a function ? Justify.
 - (iii) Determine domain and range of R .

27. (a) Find k so that

$$f(x) = \begin{cases} \frac{x^2 - 2x - 3}{x + 1}, & x \neq -1 \\ k, & x = -1 \end{cases}$$

is continuous at $x = -1$.

OR

- (b) Check the differentiability of function $f(x) = x|x|$ at $x = 0$.



28. मान ज्ञात कीजिए :

$$\int_{\pi/2}^{\pi} e^x \left(\frac{1 - \sin x}{1 - \cos x} \right) dx$$

29. (क) तीन बच्चों वाले परिवारों में लड़कों की संख्या का प्रायिकता बंटन ज्ञात कीजिए, यह मानते हुए कि लड़का व लड़की के होने की प्रायिकता समान है।

अथवा

(ख) एक सिक्का दो बार उछाला गया। माना एक यादृच्छिक चर X = चितों की संख्या – पटों की संख्या द्वारा परिभाषित किया गया है। X का प्रायिकता बंटन ज्ञात कीजिए तथा इसका माध्य भी ज्ञात कीजिए।

30. रेखाओं $\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4}$ तथा $\frac{x-4}{5} = \frac{y-1}{2} = z$ के प्रतिच्छेदन बिंदु से बिंदु $(-1, -5, -10)$ की दूरी ज्ञात कीजिए।

31. ग्राफीय विधि द्वारा निम्नलिखित रैखिक प्रोग्रामन समस्या हल कीजिए :
निम्न व्यवरोधों

$$3x + y \leq 600$$

$$x + y \leq 300$$

$$y \leq x + 200$$

$$x \geq 0, y \geq 0$$

के अंतर्गत $Z = 100x + 50y$ का अधिकतमीकरण करना।

खण्ड घ

इस खण्ड में 4 दीर्घ-उत्तरीय (LA) प्रकार के प्रश्न हैं, जिनमें प्रत्येक के 5 अंक हैं।

32. यदि A एक 3×3 व्युत्क्रमणीय आव्यूह है, तो दर्शाइए कि किसी अदिश $k \neq 0$ के लिए $(kA)^{-1} = \frac{1}{k} A^{-1}$. अतः $(3A)^{-1}$ परिकलित कीजिए, जहाँ

$$A = \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix} \text{ है।}$$



28. Evaluate :

$$\int_{\pi/2}^{\pi} e^x \left(\frac{1 - \sin x}{1 - \cos x} \right) dx$$

29. (a) Find the probability distribution of the number of boys in families having three children, assuming equal probability for a boy and a girl.

OR

- (b) A coin is tossed twice. Let X be a random variable defined as number of heads minus number of tails. Obtain the probability distribution of X and also find its mean.

30. Find the distance of the point $(-1, -5, -10)$ from the point of intersection of the lines $\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4}$ and $\frac{x-4}{5} = \frac{y-1}{2} = z$.

31. Solve the following Linear Programming Problem using graphical method :
Maximise $Z = 100x + 50y$
subject to the constraints

$$3x + y \leq 600$$

$$x + y \leq 300$$

$$y \leq x + 200$$

$$x \geq 0, y \geq 0$$

SECTION D

This section comprises 4 Long Answer (LA) type questions of 5 marks each.

32. If A is a 3×3 invertible matrix, show that for any scalar $k \neq 0$, $(kA)^{-1} = \frac{1}{k}A^{-1}$. Hence calculate $(3A)^{-1}$, where

$$A = \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix}.$$



33. सूर्य के प्रकाश के संपर्क में आने के सापेक्ष पौधे की ऊँचाई (y सेमी) के बीच का संबंध समीकरण $y = 4x - \frac{1}{2}x^2$ द्वारा नियंत्रित होता है, जहाँ x सूर्य के प्रकाश के संपर्क में आने वाले दिनों की संख्या है।

- (i) सूर्य के प्रकाश के सापेक्ष पौधे की वृद्धि दर ज्ञात कीजिए। 2
- (ii) कितने दिनों में पौधा अपनी अधिकतम ऊँचाई प्राप्त कर लेगा ? अधिकतम ऊँचाई कितनी है ? 3

34. (क) ज्ञात कीजिए :

$$\int \frac{\cos x}{(4 + \sin^2 x)(5 - 4 \cos^2 x)} dx$$

अथवा

(ख) मान ज्ञात कीजिए :

$$\int_0^{\pi} \frac{dx}{a^2 \cos^2 x + b^2 \sin^2 x}$$

35. (क) दर्शाइए कि एक समांतर चतुर्भुज जिसके विकर्ण $\vec{a}$ तथा $\vec{b}$ द्वारा निरूपित हों, का क्षेत्रफल $\frac{1}{2} |\vec{a} \times \vec{b}|$ द्वारा प्रदत्त होता है। अतः उस समांतर चतुर्भुज का क्षेत्रफल ज्ञात कीजिए जिसके विकर्ण $2\hat{i} - \hat{j} + \hat{k}$ तथा $\hat{i} + 3\hat{j} - \hat{k}$ द्वारा प्रदत्त हैं।

अथवा

(ख) एक रेखा का सदिश तथा कार्तीय समीकरण ज्ञात कीजिए जो बिंदु $(1, 2, -4)$ से होकर जाती है तथा दोनों रेखाओं $\frac{x-8}{3} = \frac{y+19}{-16} = \frac{z-10}{7}$, तथा $\vec{r} = 15\hat{i} + 29\hat{j} + 5\hat{k} + \mu(3\hat{i} + 8\hat{j} - 5\hat{k})$ के लंबवत है।



- 33.** The relation between the height of the plant (y cm) with respect to exposure to sunlight is governed by the equation $y = 4x - \frac{1}{2}x^2$, where x is the number of days exposed to sunlight.

- (i) Find the rate of growth of the plant with respect to sunlight. 2
- (ii) In how many days will the plant attain its maximum height ?
What is the maximum height ? 3

- 34.** (a) Find :

$$\int \frac{\cos x}{(4 + \sin^2 x)(5 - 4 \cos^2 x)} dx$$

OR

- (b) Evaluate :

$$\int_0^{\pi} \frac{dx}{a^2 \cos^2 x + b^2 \sin^2 x}$$

- 35.** (a) Show that the area of a parallelogram whose diagonals are represented by $\vec{a}$ and $\vec{b}$ is given by $\frac{1}{2} |\vec{a} \times \vec{b}|$. Also find the area of a parallelogram whose diagonals are $2\hat{i} - \hat{j} + \hat{k}$ and $\hat{i} + 3\hat{j} - \hat{k}$.

OR

- (b) Find the equation of a line in vector and cartesian form which passes through the point $(1, 2, -4)$ and is perpendicular to the lines $\frac{x-8}{3} = \frac{y+19}{-16} = \frac{z-10}{7}$, and

$$\vec{r} = 15\hat{i} + 29\hat{j} + 5\hat{k} + \mu(3\hat{i} + 8\hat{j} - 5\hat{k}).$$



खण्ड ड

इस खण्ड में 3 प्रकरण-अध्ययन आधारित प्रश्न हैं, जिनमें प्रत्येक के 4 अंक हैं।

प्रकरण अध्ययन – 1

- 36.** कुछ छात्रों को दशमलवों की तुलना करते समय एक ग़लत धारणा होती है। उदाहरण के लिए, कोई छात्र $78.56 > 78.9$ का उल्लेख कर सकता है क्योंकि $7856 > 789$ है। इस अवधारणा का आकलन करने के लिए, अध्यापक द्वारा कक्षा VI के विद्यार्थियों को दशमलव तुलना परीक्षण के लिए निम्नलिखित प्रश्न दिया गया :

हाल ही में स्कूल में आयोजित खेल दिवस में, 5 छात्रों ने भाला फेंक प्रतियोगिता में भाग लिया। उन्होंने जितनी दूरी तक भाला फेंका है, वह नीचे की तालिका में दर्शाया गया है :

छात्र का नाम	भाले की दूरी (मीटर में)
अजय	47.7
बिजॉय	47.07
कार्तिक	43.09
दिनेश	43.9
देवेश	45.2

विद्यार्थियों से पूछा गया कि वे बताएँ कि भाला सबसे दूर किसने फेंका है।

इस परीक्षा के आधार पर अध्यापक ने यह परिणाम प्राप्त किया कि 40% विद्यार्थियों को दशमलवों की तुलना में ग़लत धारणा है तथा शेष विद्यार्थियों को यह ग़लत धारणा नहीं है। परीक्षा में जिन विद्यार्थियों को यह ग़लत धारणा है, उनमें से 80% विद्यार्थियों ने बिजॉय को सही उत्तर के रूप में दिया जबकि जिन विद्यार्थियों को यह ग़लत धारणा नहीं है, उनमें से 90% ने अपना उत्तर बिजॉय नहीं दिया।

उपर्युक्त सूचना के आधार पर निम्नलिखित प्रश्नों के उत्तर दीजिए :

- परीक्षा में एक विद्यार्थी जिसको यह ग़लत धारणा नहीं है, के बिजॉय, सही उत्तर के रूप में देने की प्रायिकता क्या है ? 1
- इसकी क्या प्रायिकता है कि एक यादृच्छया चुना गया विद्यार्थी, परीक्षा में उत्तर के रूप में बिजॉय लिखता है ? 1
- (क) एक विद्यार्थी जिसने बिजॉय उत्तर के रूप में दिया, की ग़लत धारणा होने की क्या प्रायिकता है ? 2

अथवा

- (ख) इसकी क्या प्रायिकता है कि एक विद्यार्थी जिसने अपना उत्तर बिजॉय दिया है, उन विद्यार्थियों में से है जिसे यह ग़लत धारणा नहीं है ? 2



SECTION E

This section comprises 3 case study based questions of 4 marks each.

Case Study – 1

- 36.** Some students are having a misconception while comparing decimals. For example, a student may mention that $78.56 > 78.9$ as $7856 > 789$. In order to assess this concept, a decimal comparison test was administered to the students of class VI through the following question : In the recently held Sports Day in the school, 5 students participated in a javelin throw competition. The distances to which they have thrown the javelin are shown below in the table :

Name of student	Distance of javelin (in meters)
Ajay	47.7
Bijoy	47.07
Kartik	43.09
Dinesh	43.9
Devesh	45.2

The students were asked to identify who has thrown the javelin the farthest.

Based on the test attempted by the students, the teacher concludes that 40% of the students have the misconception in the concept of decimal comparison and the rest do not have the misconception. 80% of the students having misconception answered Bijoy as the correct answer in the paper. 90% of the students who are identified with not having misconception, did not answer Bijoy as their answer.

On the basis of the above information, answer the following questions :

- (i) What is the probability of a student not having misconception but still answers Bijoy in the test ? 1
- (ii) What is the probability that a randomly selected student answers Bijoy as his answer in the test ? 1
- (iii) (a) What is the probability that a student who answered as Bijoy is having misconception ? 2

OR

- (iii) (b) What is the probability that a student who answered as Bijoy is amongst students who do not have the misconception ? 2



प्रकरण अध्ययन – 2

37. एक इंजीनियर एक शहर में नया मेट्रो रेल नेटवर्क डिज़ाइन कर रहा है।



शुरू में, वह दो मेट्रो लाइनों A तथा B, विभिन्न स्टेशनों के लिए डिज़ाइन करता है। मेट्रो लाइन A का ट्रैक

$$l_1 : \frac{x-2}{3} = \frac{y+1}{-2} = \frac{z-3}{4}, \text{ जबकि लाइन B का ट्रैक}$$

$$l_2 : \frac{x-1}{2} = \frac{y-3}{1} = \frac{z+2}{-3} \text{ से निरूपित किया गया है।}$$

उपर्युक्त सूचना पर आधारित, निम्नलिखित प्रश्नों के उत्तर दीजिए :

- (i) ज्ञात कीजिए कि क्या दो मेट्रो ट्रैक परस्पर समांतर हैं। 1
- (ii) मेट्रो स्टेशनों की छतों पर सोलर पैनल लगाए जाने हैं। मेट्रो लाइन A के स्टेशनों की छतों पर, लाइन A के ट्रैक (l_1) के समांतर तथा बिंदु $(1, -2, -3)$ से हो कर जाने वाले सौर पैनलों को निरूपित करने वाली रेखा का समीकरण ज्ञात कीजिए। 1
- (iii) (क) स्टेशनों को जोड़ने के लिए, दोनों मेट्रो लाइनों के लंबवत एक पैदल यात्री पथ बनाना है जो बिंदु $(3, 2, 1)$ से हो कर जाए। इस पैदल यात्री पथ का समीकरण ज्ञात कीजिए। 2

अथवा

- (iii) (ख) लाइन A तथा लाइन B के बीच की न्यूनतम दूरी ज्ञात कीजिए। 2



Case Study – 2

37. An engineer is designing a new metro rail network in a city.



Initially, two metro lines, Line A and Line B, each consisting of multiple stations are designed. The track for Line A is represented by

$$l_1 : \frac{x-2}{3} = \frac{y+1}{-2} = \frac{z-3}{4}, \text{ while the track for Line B is represented by } l_2 : \frac{x-1}{2} = \frac{y-3}{1} = \frac{z+2}{-3}.$$

Based on the above information, answer the following questions :

- (i) Find whether the two metro tracks are parallel. 1
- (ii) Solar panels are to be installed on the rooftop of the metro stations. Determine the equation of the line representing the placement of solar panels on the rooftop of Line A's stations, given that panels are to be positioned parallel to Line A's track (l_1) and pass through the point $(1, -2, -3)$. 1
- (iii) (a) To connect the stations, a pedestrian pathway perpendicular to the two metro lines is to be constructed which passes through point $(3, 2, 1)$. Determine the equation of the pedestrian walkway. 2

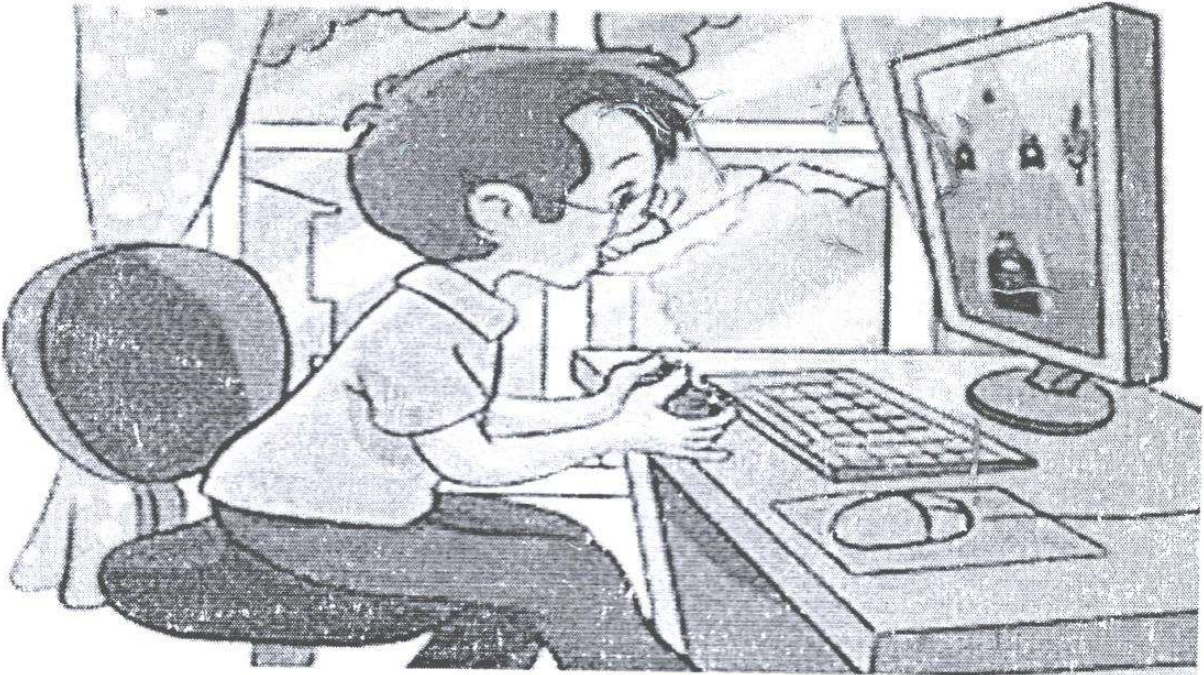
OR

- (iii) (b) Find the shortest distance between Line A and Line B. 2



प्रकरण अध्ययन – 3

38. भारी गेमिंग सत्र के दौरान, एक छात्र के लैपटॉप प्रोसेसर का तापमान काफी बढ़ जाता है। सत्र के बाद में यह ठंडा होने लगता है तथा इसके ठंडा होने की दर, प्रोसेसर के तापमान तथा कमरे के तापमान (25°C) के अंतर के समानुपाती है। आरंभ में प्रोसेसर का तापमान 85°C है तथा ठंडा होने की दर समीकरण $\frac{d}{dt}(T(t)) = -k(T(t) - 25)$ द्वारा परिभाषित है, जहाँ $T(t)$, समय t (मिनट में) पर प्रोसेसर के तापमान को निरूपित करता है तथा k एक अचर है।



उपर्युक्त सूचना पर आधारित, निम्नलिखित प्रश्नों के उत्तर दीजिए :

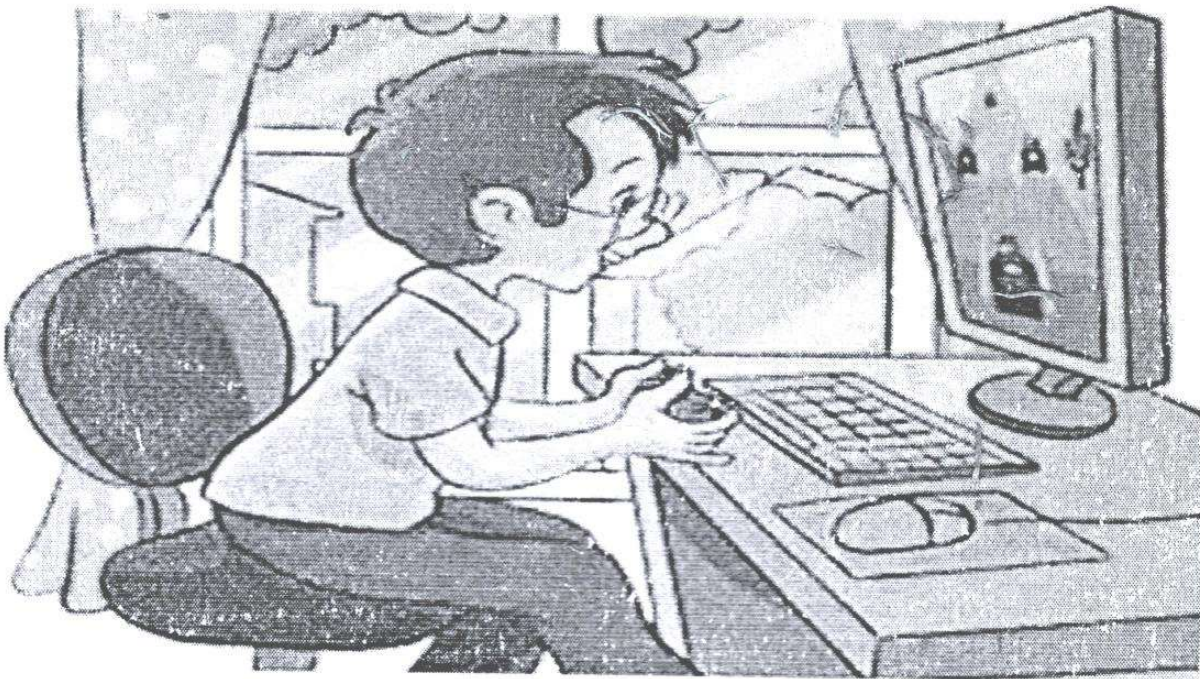
- (i) प्रोसेसर के तापमान $T(t)$ के लिए व्यंजक ज्ञात कीजिए जबकि दिया गया है कि $T(0) = 85^{\circ}\text{C}$ है। 2
- (ii) प्रोसेसर के तापमान को 40°C तक आने में कितना समय लगेगा ? दिया गया है कि $k = 0.03$, $\log_e 4 = 1.3863$. 2



Case Study – 3

38. During a heavy gaming session, the temperature of a student's laptop processor increases significantly. After the session, the processor begins to cool down, and the rate of cooling is proportional to the difference between the processor's temperature and the room temperature (25°C). Initially the processor's temperature is 85°C . The rate of cooling is defined by the equation $\frac{d}{dt}(T(t)) = -k(T(t) - 25)$,

where $T(t)$ represents the temperature of the processor at time t (in minutes) and k is a constant.



Based on the above information, answer the following questions :

- (i) Find the expression for temperature of processor, $T(t)$ given that $T(0) = 85^{\circ}\text{C}$. 2
- (ii) How long will it take for the processor's temperature to reach 40°C ? Given that $k = 0.03$, $\log_e 4 = 1.3863$. 2

Marking Scheme
Strictly Confidential
(For Internal and Restricted use only)
Senior School Examination, 2025
SUBJECT: MATHEMATICS (Q.P. CODE – 65/4/1)

General Instructions: -

1	You are aware that evaluation is the most important process in the actual and correct assessment of the candidates. A small mistake in evaluation may lead to serious problems which may affect the future of the candidates, education system and teaching profession. To avoid mistakes, it is requested that before starting evaluation, you must read and understand the spot evaluation guidelines carefully.
2	“Evaluation policy is a confidential policy as it is related to the confidentiality of the examinations conducted, Evaluation done and several other aspects. Its leakage to the public in any manner could lead to derailment of the examination system and affect the life and future of millions of candidates. Sharing this policy/document to anyone, publishing in any magazine and printing in Newspaper/Website, etc. may invite action under various rules of the Board and IPC.”
3	Evaluation is to be done as per instructions provided in the Marking Scheme. It should not be done according to one’s own interpretation or any other consideration. The Marking Scheme should be strictly adhered to and religiously followed. However, while evaluating, answers which are based on latest information or knowledge and/or are innovative, they may be assessed for their correctness otherwise and due marks be awarded to them. In class-XII, while evaluating the competency-based questions, please try to understand the given answer and even if reply is not from a marking scheme but correct competency is enumerated by the candidate, due marks should be awarded.
4	The Marking Scheme carries only suggested value points for the answers. These are Guidelines only and do not constitute the complete answer. The students can have their own expression and if the expression is correct, the due marks should be awarded accordingly.
5	The Head-Examiner must go through the first five answer books evaluated by each evaluator on the first day, to ensure that evaluation has been carried out as per the instructions given in the Marking Scheme. If there is any variation, the same should be zero after deliberation and discussion. The remaining answer books meant for evaluation shall be given only after ensuring that there is no significant variation in the marking of individual evaluators.
6	Evaluators will mark (✓) wherever answer is correct. For wrong answer CROSS ‘X’ be marked. Evaluators will not put right (✓) while evaluating which gives the impression that the answer is correct, and no marks are awarded. This is the most common mistake which evaluators are committing.
7	If a question has parts, please award marks on the right-hand side for each part. Marks awarded for different parts of the question should then be totaled up and written in the left-hand margin and encircled. This may be followed strictly.
8	If a question does not have any parts, marks must be awarded in the left-hand margin and encircled. This may also be followed strictly.
9	If a student has attempted an extra question, answer to the question deserving more marks should be retained and the other answer scored out with a note “Extra Question” .

10	No marks to be deducted for the cumulative effect of an error. It should be penalized only once.
11	A full scale of marks_(example 0 to 80/70/60/50/40/30 marks as given in Question Paper) has to be used. Please do not hesitate to award full marks if the answer deserves it.
12	Every examiner must necessarily do evaluation work for full working hours, i.e., 8 hours every day and evaluate 20 answer books per day in main subjects and 25 answer books per day in other subjects (Details are given in Spot Guidelines). This is in view of the reduced syllabus and number of questions in question paper.
13	Ensure that you do not make the following common types of errors committed by the Examiner in the past: - • Leaving answer or part thereof unassessed in an answer book. • Giving more marks for an answer than assigned to it. • Wrong totaling of marks awarded on an answer. • Wrong transfer of marks from the inside pages of the answer book to the title page. • Wrong question wise totaling on the title page. • Wrong totaling of marks of the two columns on the title page. • Wrong grand total. • Marks in words and figures not tallying/not same. • Wrong transfer of marks from the answer book to online award list. • Answers marked as correct, but marks not awarded. (Ensure that the right tick mark is correctly and clearly indicated. It should merely be a line. Same is with the X for incorrect answer.) • Half or a part of the answer marked correct and the rest as wrong, but no marks
14	While evaluating the answer books if the answer is found to be totally incorrect, it should be marked as cross (X) and awarded zero (0) Marks.
15	Any unassessed portion, non-carrying over of marks to the title page, or total error detected by the candidate shall damage the prestige of all the personnel engaged in the evaluation work as also of the Board. Hence, to uphold the prestige of all concerned, it is again reiterated that the instructions be followed meticulously and judiciously.
16	The Examiners should acquaint themselves with the guidelines given in the “ Guidelines for Spot Evaluation ” before starting the actual evaluation.
17	Every Examiner shall also ensure that all the answers are evaluated, marks carried over to the title page, correctly totaled and written in figures and words.
18	The candidates are entitled to obtain a photocopy of the Answer Book on request on payment of the prescribed processing fee. All Examiners/Additional Head Examiners/Head Examiners are once again reminded that they must ensure that evaluation is carried out strictly as per value points for each answer as given in the Marking Scheme.

MARKING SCHEME
SENIOR SECONDARY SCHOOL EXAMINATION 2024-25
MATHEMATICS (Code-041)
[Paper Code: 65/4/1]

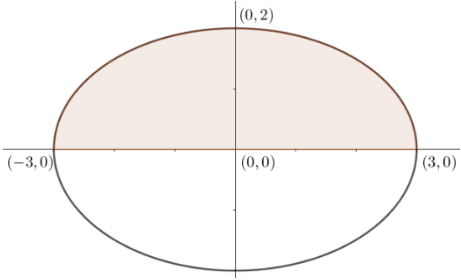
Q. No.	EXPECTED ANSWER / VALUE POINTS	Marks
	SECTION - A	
	Questions no. 1 to 18 are multiple choice questions (MCQs) of 1 mark each .	
Q1.	The principal value of $\sin^{-1}\left(\sin\left(-\frac{10\pi}{3}\right)\right)$ is : (A) $-\frac{2\pi}{3}$ (B) $-\frac{\pi}{3}$ (C) $\frac{\pi}{3}$ (D) $\frac{2\pi}{3}$	
A1.	(C) $\frac{\pi}{3}$	1
Q2.	If A and B are square matrices of same order such that $AB = A$ and $BA = B$, then $A^2 + B^2$ is equal to : (A) $A + B$ (B) BA (C) $2(A + B)$ (D) $2BA$	
A2.	(A) $A + B$	1
Q3.	For real x, let $f(x) = x^3 + 5x + 1$. Then : (A) f is one-one but not onto on R (B) f is onto on R but not one-one (C) f is one-one and onto on R (D) f is neither one-one nor onto on R	
A3.	(C) f is one-one and onto on R	1
Q4.	If $y = \sin^{-1} x$, then $(1 - x^2)\frac{d^2y}{dx^2}$ is equal to : (A) $x\frac{dy}{dx}$ (B) $-x\frac{dy}{dx}$ (C) $x^2\frac{dy}{dx}$ (D) $-x^2\frac{dy}{dx}$	
A4.	(A) $x\frac{dy}{dx}$	1

Q5.	The values of λ so that $f(x) = \sin x - \cos x - \lambda x + C$ decreases for all real values of x are : (A) $1 < \lambda < \sqrt{2}$ (B) $\lambda \geq 1$ (C) $\lambda \geq \sqrt{2}$ (D) $\lambda < 1$	
A5.	(C) $\lambda \geq \sqrt{2}$	1
Q6.	If P is a point on the line segment joining (3, 6, -1) and (6, 2, -2) and y-coordinate of P is 4, then its z-coordinate is : (A) $-\frac{3}{2}$ (B) 0 (C) 1 (D) $\frac{3}{2}$	
A6.	(A) $-\frac{3}{2}$	1
Q7.	If M and N are square matrices of order 3 such that $\det(M) = m$ and $MN = mI$, then $\det(N)$ is equal to : (A) -1 (B) 1 (C) $-m^2$ (D) m^2	
A7.	(D) m^2	1
Q8.	If $f(x) = \begin{cases} 3x - 2, & 0 < x \leq 1 \\ 2x^2 + ax, & 1 < x < 2 \end{cases}$ is continuous for $x \in (0, 2)$, then a is equal to : (A) -4 (B) $-\frac{7}{2}$ (C) -2 (D) -1	
A8.	(D) -1	1

Q13.	If $f(x) = 2x + \cos x$, then $f(x)$: (A) has a maxima at $x = \pi$ (B) has a minima at $x = \pi$ (C) is an increasing function (D) is a decreasing function	
A13.	(C) is an increasing function	1
Q14.	$\int \frac{\cos 2x - \cos 2\alpha}{\cos x - \cos \alpha} dx$ is equal to : (A) $2(\sin x + x \cos \alpha) + C$ (B) $2(\sin x - x \cos \alpha) + C$ (C) $2(\sin x + 2x \cos \alpha) + C$ (D) $2(\sin x + \sin \alpha) + C$	
A14.	(A) $2(\sin x + x \cos \alpha) + C$	1
Q15.	The value of $\int_0^1 \frac{dx}{e^x + e^{-x}}$ is : (A) $-\frac{\pi}{4}$ (B) $\frac{\pi}{4}$ (C) $\tan^{-1} e - \frac{\pi}{4}$ (D) $\tan^{-1} e$	
A15.	(C) $\tan^{-1} e - \frac{\pi}{4}$	1
Q16.	The order and degree of the differential equation $\left(\frac{d^2y}{dx^2}\right)^2 + \left(\frac{dy}{dx}\right)^2 = x \sin\left(\frac{dy}{dx}\right)$ are : (A) order 2, degree 2 (B) order 2, degree 1 (C) order 2, degree not defined (D) order 1, degree not defined	
A16.	(C) order 2, degree not defined	1
Q17.	The area of the region enclosed by the curve $y = \sqrt{x}$ and the lines $x = 0$ and $x = 4$ and x-axis is : (A) $\frac{16}{9}$ sq. units (B) $\frac{32}{9}$ sq. units (C) $\frac{16}{3}$ sq. units (D) $\frac{32}{3}$ sq. units	
A17.	(C) $\frac{16}{3}$ sq. units	1

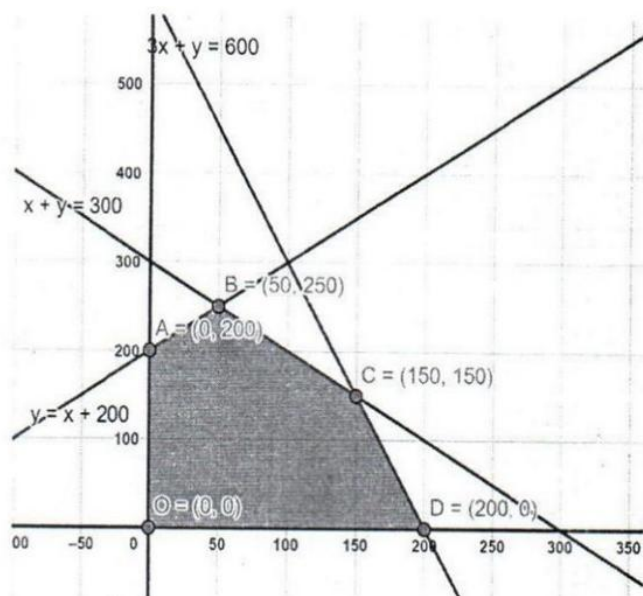
Q18.	The corner points of the feasible region of a Linear Programming Problem are (0, 2), (3, 0), (6, 0), (6, 8) and (0, 5). If $Z = ax + by$; ($a, b > 0$) be the objective function, and maximum value of Z is obtained at (0, 2) and (3, 0), then the relation between a and b is : (A) $a = b$ (B) $a = 3b$ (C) $b = 6a$ (D) $3a = 2b$	
A18.	(D) $3a = 2b$	1
Questions number 19 and 20 are Assertion and Reason based questions. Two statements are given, one labelled Assertion (A) and the other labelled Reason (R). Select the correct answer from the codes (A), (B), (C) and (D) as given below. (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A). (B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (A). (C) Assertion (A) is true, but Reason (R) is false. (D) Assertion (A) is false, but Reason (R) is true.		
Q19.	Assertion (A) : If A and B are two events such that $P(A \cap B) = 0$, then A and B are independent events. Reason (R) : Two events are independent if the occurrence of one does not effect the occurrence of the other.	
A19.	(D) Assertion (A) is false, but Reason (R) is true.	1
Q20.	Assertion (A) : In a Linear Programming Problem, if the feasible region is empty, then the Linear Programming Problem has no solution. Reason (R) : A feasible region is defined as the region that satisfies all the constraints.	
A20.	(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (A).	1
SECTION B This section comprises very short answer (VSA) type questions of 2 marks each.		
Q21.	Let A and B be two square matrices of order 3 such that $\det(A) = 3$ and $\det(B) = -4$. Find the value of $\det(-6AB)$.	
A21.	$ -6AB = (-6)^3 A B $ $= -216 \times 3 \times (-4) = 2592$	1 1

Q22.	(a) Find the least value of 'a' so that $f(x) = 2x^2 - ax + 3$ is an increasing function on $[2, 4]$. OR (b) If $f(x) = x + \frac{1}{x}$, $x \geq 1$, show that f is an increasing function.	
A22.(a)	$f(x) = 2x^2 - ax + 3 \Rightarrow f'(x) = 4x - a$ Now $2 \leq x \leq 4 \Rightarrow 8 - a \leq 4x - a \leq 16 - a$ For f to be an increasing function, $f'(x) \geq 0$ $\Rightarrow 8 - a \geq 0 \Rightarrow a \leq 8$ $\therefore$ Least value of a does not exist.	$\frac{1}{2}$ 1 $\frac{1}{2}$
OR		
A22.(b)	$f(x) = x + \frac{1}{x} \Rightarrow f'(x) = 1 - \frac{1}{x^2} = \frac{x^2 - 1}{x^2}$ Now $\frac{x^2 - 1}{x^2} \geq 0$ for all $x \geq 1$ $\Rightarrow f'(x) \geq 0 \Rightarrow f$ is an increasing function.	1 $\frac{1}{2}$ $\frac{1}{2}$
Q23.	(a) Simplify $\sin^{-1} \left(\frac{x}{\sqrt{1+x^2}} \right)$. OR (b) Find domain of $\sin^{-1} \sqrt{x-1}$.	
A23.(a)	Put $x = \tan \theta \Rightarrow \theta = \tan^{-1} x$ Now $\sin^{-1} \left(\frac{x}{\sqrt{1+x^2}} \right)$ $= \sin^{-1} \left(\frac{\tan \theta}{\sec \theta} \right) = \sin^{-1} (\sin \theta)$ $= \theta = \tan^{-1} x$	$\frac{1}{2}$ 1 $\frac{1}{2}$
OR		
A23.(b)	Here $-1 \leq \sqrt{x-1} \leq 1$ $\Rightarrow 0 \leq x-1 \leq 1 \Rightarrow 1 \leq x \leq 2$ Hence, domain is $x \in [1, 2]$	1 1

Q24.	Calculate the area of the region bounded by the curve $\frac{x^2}{9} + \frac{y^2}{4} = 1$ and the x-axis using integration.		
A24.		$A = 2 \times \frac{2}{3} \int_0^3 \sqrt{9-x^2} dx$ $= \frac{4}{3} \left[\frac{x}{2} \sqrt{9-x^2} + \frac{9}{2} \sin^{-1} \left(\frac{x}{3} \right) \right]_0^3$ $= \frac{4}{3} \left[\left(0 + \frac{9}{2} \sin^{-1} 1 \right) - 0 \right]$ $= 3\pi$	(½ for correct figure) ½ ½ ½
Q25.	For the curve $y = 5x - 2x^3$, if x increases at the rate of 2 units/s, then how fast is the slope of the curve changing when $x = 2$?		
A25.	$y = 5x - 2x^3$ Given $\frac{dx}{dt} = 2 \text{ units/s}$ slope of the curve $= \frac{dy}{dx} = 5 - 6x^2 = m$ $\frac{dm}{dt} = -12x \frac{dx}{dt} = -12x(2) = -24x$ at $x = 2$, $\frac{dm}{dt} = -24(2) = -48$ Hence, slope of curve is decreasing at the rate of 48		½ ½ ½ ½
SECTION C			
This section comprises short answer (SA) type questions of 3 marks each .			
Q26.	(a) If $f : \mathbb{R}^+ \rightarrow \mathbb{R}$ is defined as $f(x) = \log_a x$ ($a > 0$ and $a \neq 1$), prove that f is a bijection. $(\mathbb{R}^+ \text{ is a set of all positive real numbers.})$ OR (b) Let $A = \{1, 2, 3\}$ and $B = \{4, 5, 6\}$. A relation R from A to B is defined as $R = \{(x, y) : x + y = 6, x \in A, y \in B\}$. (i) Write all elements of R. (ii) Is R a function ? Justify. (iii) Determine domain and range of R.		

A29.(b)	Possible values of X are $-2, 0, 2$ <table><tr><td>X</td><td>-2</td><td>0</td><td>2</td></tr><tr><td>$P(X)$</td><td>$\frac{1}{4}$</td><td>$\frac{2}{4} = \frac{1}{2}$</td><td>$\frac{1}{4}$</td></tr></table> Mean = $\sum XP(X) = -2\left(\frac{1}{4}\right) + 0\left(\frac{1}{2}\right) + 2\left(\frac{1}{4}\right) = 0$	X	-2	0	2	$P(X)$	$\frac{1}{4}$	$\frac{2}{4} = \frac{1}{2}$	$\frac{1}{4}$	$\frac{1}{2}$ $1\frac{1}{2}$ 1
X	-2	0	2							
$P(X)$	$\frac{1}{4}$	$\frac{2}{4} = \frac{1}{2}$	$\frac{1}{4}$							
Q30.	Find the distance of the point $(-1, -5, -10)$ from the point of intersection of the lines $\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4}$ and $\frac{x-4}{5} = \frac{y-1}{2} = z$.									
A30.	$l_1: \frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4} = \lambda$ Any point on l_1 is $(2\lambda + 1, 3\lambda + 2, 4\lambda + 3)$ $l_2: \frac{x-4}{5} = \frac{y-1}{2} = \frac{z-0}{1} = \mu$ Any point on l_2 is $(5\mu + 4, 2\mu + 1, \mu)$ For point of intersection, $2\lambda + 1 = 5\mu + 4, 3\lambda + 2 = 2\mu + 1$ Solving, $\lambda = \mu = -1$ Since, $\lambda = \mu = -1$ satisfy $4\lambda + 3 = \mu$ $\therefore$ Point of intersection is $(-1, -1, -1)$ Now distance of $(-1, -5, -10)$ from $(-1, -1, -1)$ is: $\sqrt{(-1+1)^2 + (-1+5)^2 + (-1+10)^2} = \sqrt{97}$ units	1 1 $\frac{1}{2}$ $\frac{1}{2}$								
Q31.	Solve the following Linear Programming Problem using graphical method : Maximise $Z = 100x + 50y$ subject to the constraints $3x + y \leq 600$ $x + y \leq 300$ $y \leq x + 200$ $x \geq 0, y \geq 0$									

A31.



Corner Point	Value of $Z = 100x + 50y$
$O(0,0)$	0
$A(0,200)$	10000
$B(50,250)$	17500
$C(150,150)$	22500
$D(200,0)$	20000

$$Z_{\max} = 22500 \text{ when } x = 150, y = 150$$

For correct graph and shading
 $1\frac{1}{2}$ For correct table
1 $\frac{1}{2}$ **SECTION D**This section comprises long answer (LA) type questions of **5 marks each**.**Q32.**

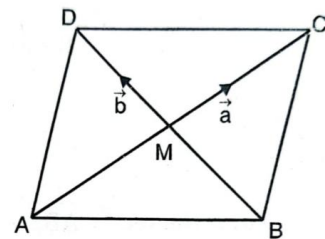
If A is a 3×3 invertible matrix, show that for any scalar $k \neq 0$, $(kA)^{-1} = \frac{1}{k}A^{-1}$. Hence calculate $(3A)^{-1}$, where

$$A = \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix}.$$

A32.	Consider $(kA)\left(\frac{1}{k}A^{-1}\right) = k \cdot \frac{1}{k}(A \cdot A^{-1}) = I$ $\Rightarrow kA$ and $\frac{1}{k}A^{-1}$ are inverse of each other. $\therefore (kA)^{-1} = \frac{1}{k}A^{-1}$ $\therefore (3A)^{-1} = \frac{1}{3}A^{-1}$ Here, $A = 4 \neq 0 \therefore A^{-1}$ exists. $adjA = \begin{bmatrix} 3 & 1 & -1 \\ 1 & 3 & 1 \\ -1 & 1 & 3 \end{bmatrix}$ $\therefore A^{-1} = \frac{1}{ A } \cdot adjA = \frac{1}{4} \begin{bmatrix} 3 & 1 & -1 \\ 1 & 3 & 1 \\ -1 & 1 & 3 \end{bmatrix}$ $\therefore (3A)^{-1} = \frac{1}{12} \begin{bmatrix} 3 & 1 & -1 \\ 1 & 3 & 1 \\ -1 & 1 & 3 \end{bmatrix}$	1 1 2 $\frac{1}{2}$ $\frac{1}{2}$
Q33.	The relation between the height of the plant (y cm) with respect to exposure to sunlight is governed by the equation $y = 4x - \frac{1}{2}x^2$, where x is the number of days exposed to sunlight. (i) Find the rate of growth of the plant with respect to sunlight. (ii) In how many days will the plant attain its maximum height ? What is the maximum height ?	2 3
A33.	(i) $y = 4x - \frac{1}{2}x^2 \Rightarrow \frac{dy}{dx} = (4 - x)$ cm/day (ii) For maximum height, $\frac{dy}{dx} = 0 \Rightarrow x = 4$ days as $\frac{d^2y}{dx^2} < 0$, number of days = 4 Now, Maximum height = $y(4) = 16 - \frac{1}{2}(16) = 8$ cm	2 1 1 1

Q34.	(a) Find : $\int \frac{\cos x}{(4 + \sin^2 x)(5 - 4 \cos^2 x)} dx$ OR (b) Evaluate : $\int_0^{\pi} \frac{dx}{a^2 \cos^2 x + b^2 \sin^2 x}$
A34.(a)	$I = \int \frac{\cos x}{(4 + \sin^2 x)(5 - 4 \cos^2 x)} dx$ $= \int \frac{\cos x}{(4 + \sin^2 x)(1 + 4 \sin^2 x)} dx$ $\sin x = t$ gives $I = \int \frac{dt}{(4 + t^2)(1 + 4t^2)}$ $= -\frac{1}{15} \int \frac{dt}{4 + t^2} + \frac{4}{15} \int \frac{dt}{1 + 4t^2} \quad (\because \text{using Partial Fraction})$ $= -\frac{1}{30} \tan^{-1}\left(\frac{t}{2}\right) + \frac{2}{15} \tan^{-1}(2t) + C$ $= -\frac{1}{30} \tan^{-1}\left(\frac{\sin x}{2}\right) + \frac{2}{15} \tan^{-1}(2 \sin x) + C$ 1 $\frac{1}{2}$ 2 1 $\frac{1}{2}$
OR	

A34.(b)	$I = \int_0^{\pi} \frac{dx}{a^2 \cos^2 x + b^2 \sin^2 x}$ $= 2 \int_0^{\pi/2} \frac{dx}{a^2 \cos^2 x + b^2 \sin^2 x}$ $= 2 \int_0^{\pi/2} \frac{\sec^2 x}{a^2 + b^2 \tan^2 x} dx$ $\tan x = t$ gives $I = 2 \int_0^{\infty} \frac{dt}{a^2 + b^2 t^2}$ $= \frac{2}{b^2} \cdot \frac{b}{a} \tan^{-1} \left(\frac{bt}{a} \right) \Bigg _0^{\infty}$ $= \frac{\pi}{ab}$	$\frac{1}{2}$ 1 $\frac{1}{2}$ $\frac{1}{2}$ $1\frac{1}{2}$ 1
Q35 .	(a) Show that the area of a parallelogram whose diagonals are represented by $\vec{a}$ and $\vec{b}$ is given by $\frac{1}{2} \vec{a} \times \vec{b}$. Also find the area of a parallelogram whose diagonals are $2\hat{i} - \hat{j} + \hat{k}$ and $\hat{i} + 3\hat{j} - \hat{k}$. OR (b) Find the equation of a line in vector and cartesian form which passes through the point $(1, 2, -4)$ and is perpendicular to the lines $\frac{x-8}{3} = \frac{y+19}{-16} = \frac{z-10}{7}$, and $\vec{r} = 15\hat{i} + 29\hat{j} + 5\hat{k} + \mu(3\hat{i} + 8\hat{j} - 5\hat{k}).$	
A35.(a)	Let $ABCD$ be the parallelogram with diagonals $\overrightarrow{AB} = \vec{a}$ and $\overrightarrow{BD} = \vec{b}$. $\therefore \overrightarrow{AB} = \frac{1}{2}(\vec{a} - \vec{b})$ and $\overrightarrow{AD} = \frac{1}{2}(\vec{a} + \vec{b})$ Area of $ABCD$ $= \overrightarrow{AB} \times \overrightarrow{AD} $ $= \left \frac{1}{2}(\vec{a} - \vec{b}) \times \frac{1}{2}(\vec{a} + \vec{b}) \right $ $= \frac{1}{4} \vec{a} \times \vec{a} + \vec{a} \times \vec{b} - \vec{b} \times \vec{a} - \vec{b} \times \vec{b} $	$\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$



	$= \frac{1}{4} \vec{a} \times \vec{b} + \vec{a} \times \vec{b} \quad (\because \vec{a} \times \vec{a} = \vec{0})$ $= \frac{1}{4} 2(\vec{a} \times \vec{b}) $ $= \frac{1}{2} \vec{a} \times \vec{b} $ Given $\vec{a} = 2\hat{i} - \hat{j} + \hat{k}$, $\vec{b} = \hat{i} + 3\hat{j} - \hat{k}$ Area of parallelogram $= \frac{1}{2} \vec{a} \times \vec{b}$ Now $\vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & -1 & 1 \\ 1 & 3 & -1 \end{vmatrix} = -2\hat{i} + 3\hat{j} + 7\hat{k}$ $ \vec{a} \times \vec{b} = \sqrt{62}$ Area of parallelogram $= \frac{1}{2} \sqrt{62}$	$\frac{1}{2}$ 2 $\frac{1}{2}$ $\frac{1}{2}$
OR		
A35.(b)	Given lines are $\frac{x-8}{3} = \frac{y+19}{-16} = \frac{z-10}{7}$ and $\vec{r} = (15\hat{i} + 29\hat{j} + 5\hat{k}) + \mu(3\hat{i} + 8\hat{j} - 5\hat{k})$ The first line in vector form is $\vec{r} = (8\hat{i} - 19\hat{j} + 10\hat{k}) + \lambda(3\hat{i} - 16\hat{j} + 7\hat{k})$ $\vec{a}_1 = 8\hat{i} - 19\hat{j} + 10\hat{k}$, $\vec{a}_2 = 15\hat{i} + 29\hat{j} + 5\hat{k}$ $\vec{b}_1 = 3\hat{i} - 16\hat{j} + 7\hat{k}$, $\vec{b}_2 = 3\hat{i} + 8\hat{j} - 5\hat{k}$ $\vec{b}_1 \times \vec{b}_2 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & -16 & 7 \\ 3 & 8 & -5 \end{vmatrix} = 24\hat{i} + 36\hat{j} + 72\hat{k}$ $\therefore$ Equation of line passing through $(1, 2, -4)$ and parallel to $\vec{b}$ is $\vec{r} = (\hat{i} + 2\hat{j} - 4\hat{k}) + t(24\hat{i} + 36\hat{j} + 72\hat{k})$ or $\vec{r} = (\hat{i} + 2\hat{j} - 4\hat{k}) + t'(2\hat{i} + 3\hat{j} + 6\hat{k})$ Cartesian form of line is $\frac{x-1}{24} = \frac{y-2}{36} = \frac{z+4}{72}$ or $\frac{x-1}{2} = \frac{y-2}{3} = \frac{z+4}{6}$	1 1 1 1 1

SECTION E

This section comprises 3 case study-based questions of **4 marks each**.

Q36.

Some students are having a misconception while comparing decimals. For example, a student may mention that $78.56 > 78.9$ as $7856 > 789$. In order to assess this concept, a decimal comparison test was administered to the students of class VI through the following question : In the recently held Sports Day in the school, 5 students participated in a javelin throw competition. The distances to which they have thrown the javelin are shown below in the table :

Name of student	Distance of javelin (in meters)
Ajay	47.7
Bijoy	47.07
Kartik	43.09
Dinesh	43.9
Devesh	45.2

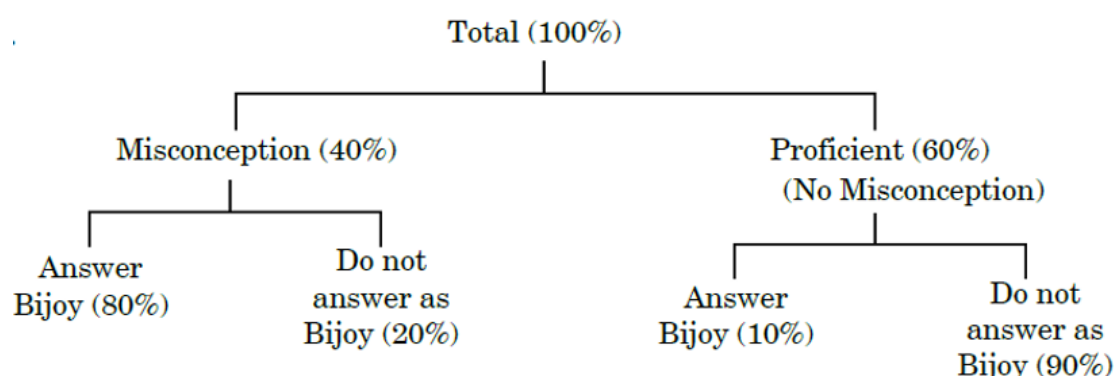
The students were asked to identify who has thrown the javelin the farthest.

Based on the test attempted by the students, the teacher concludes that 40% of the students have the misconception in the concept of decimal comparison and the rest do not have the misconception. 80% of the students having misconception answered Bijoy as the correct answer in the paper. 90% of the students who are identified with not having misconception, did not answer Bijoy as their answer.

On the basis of the above information, answer the following questions :

- | | | |
|-----------|--|---|
| (i) | What is the probability of a student not having misconception but still answers Bijoy in the test ? | 1 |
| (ii) | What is the probability that a randomly selected student answers Bijoy as his answer in the test ? | 1 |
| (iii) | (a) What is the probability that a student who answered as Bijoy is having misconception ? | 2 |
| OR | | |
| (iii) | (b) What is the probability that a student who answered as Bijoy is amongst students who do not have the misconception ? | 2 |

A36.



	Let E_1 : Student has a misconception E_2 : Student does not have misconception A: Student answered Bijoy as correct $\therefore P(E_1) = \frac{40}{100}, P(E_2) = \frac{60}{100}$ $P(A E_1) = \frac{80}{100}, P(A E_2) = \frac{10}{100}$ $P(\bar{A} E_1) = \frac{20}{100}, P(\bar{A} E_2) = \frac{90}{100}$ (i) $P(A E_2) = \frac{10}{100}$ or $\frac{1}{10}$ (ii) $P(A) = P(E_1)P(A E_1) + P(E_2)P(A E_2)$ $= \frac{40}{100} \times \frac{80}{100} + \frac{60}{100} \times \frac{10}{100}$ $= \frac{38}{100} \text{ or } \frac{19}{50}$ (iii)(a) $P(E_1 A) = \frac{P(E_1)P(A E_1)}{P(A)}$ $= \frac{\frac{40}{100} \times \frac{80}{100}}{\frac{38}{100}} = \frac{16}{19}$ OR (iii)(b) $P(E_2 A) = \frac{P(E_2)P(A E_2)}{P(A)}$ $= \frac{\frac{60}{100} \times \frac{10}{100}}{\frac{38}{100}} = \frac{3}{19}$	1 1 2 2
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Q37.

An engineer is designing a new metro rail network in a city.



Initially, two metro lines, Line A and Line B, each consisting of multiple stations are designed. The track for Line A is represented by

$$l_1 : \frac{x-2}{3} = \frac{y+1}{-2} = \frac{z-3}{4}, \text{ while the track for Line B is represented by } l_2 : \frac{x-1}{2} = \frac{y-3}{1} = \frac{z+2}{-3}.$$

Based on the above information, answer the following questions :

- (i) Find whether the two metro tracks are parallel. 1
- (ii) Solar panels are to be installed on the rooftop of the metro stations. Determine the equation of the line representing the placement of solar panels on the rooftop of Line A's stations, given that panels are to be positioned parallel to Line A's track (l_1) and pass through the point $(1, -2, -3)$. 1
- (iii) (a) To connect the stations, a pedestrian pathway perpendicular to the two metro lines is to be constructed which passes through point $(3, 2, 1)$. Determine the equation of the pedestrian walkway. 2

OR

- (iii) (b) Find the shortest distance between Line A and Line B. 2

A37.

$$l_1 : \frac{x-2}{3} = \frac{y+1}{-2} = \frac{z-3}{4} ; l_2 : \frac{x-1}{2} = \frac{y-3}{1} = \frac{z+2}{-3}$$

(i) Drs of l_1 are $\langle 3, -2, 4 \rangle$, Drs of l_2 are $\langle 2, 1, -3 \rangle$

as Drs are not proportional, hence l_1 is not parallel to l_2 .

(ii) Equations of line parallel to l_1 and passing through $(1, -2, -3)$ is

$$\frac{x-1}{3} = \frac{y+2}{-2} = \frac{z+3}{4} \text{ or } \vec{r} = (\hat{i} - 2\hat{j} - 3\hat{k}) + \lambda(3\hat{i} - 2\hat{j} + 4\hat{k})$$

1

1

	(iii)(a) The pathway is perpendicular to l_1 and l_2. $\therefore$ It is parallel to $\vec{b}_1 \times \vec{b}_2$ $\vec{b} = \vec{b}_1 \times \vec{b}_2 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & -2 & 4 \\ 2 & 1 & -3 \end{vmatrix} = 2\hat{i} + 17\hat{j} + 7\hat{k}$ $\therefore$ Equation of pathway is $\vec{r} = (3\hat{i} + 2\hat{j} + \hat{k}) + \lambda(2\hat{i} + 17\hat{j} + 7\hat{k})$ OR (iii)(b) $\vec{a}_1 = 2\hat{i} - \hat{j} + 3\hat{k}$, $\vec{a}_2 = \hat{i} + 3\hat{j} - 2\hat{k}$ $\vec{b}_1 = 3\hat{i} - 2\hat{j} + 4\hat{k}$, $\vec{b}_2 = 2\hat{i} + \hat{j} - 3\hat{k}$ $d = \frac{ (\vec{a}_2 - \vec{a}_1) \cdot (\vec{b}_1 \times \vec{b}_2) }{ \vec{b}_1 \times \vec{b}_2 }$ $= \frac{ (-\hat{i} + 4\hat{j} - 5\hat{k}) \cdot (2\hat{i} + 17\hat{j} + 7\hat{k}) }{\sqrt{4 + 289 + 49}}$ $= \frac{31}{\sqrt{342}}$	1 1 1 1
Q38.	During a heavy gaming session, the temperature of a student's laptop processor increases significantly. After the session, the processor begins to cool down, and the rate of cooling is proportional to the difference between the processor's temperature and the room temperature (25°C). Initially the processor's temperature is 85°C. The rate of cooling is defined by the equation $\frac{d}{dt}(T(t)) = -k(T(t) - 25)$, where $T(t)$ represents the temperature of the processor at time t (in minutes) and k is a constant. Based on the above information, answer the following questions : (i) Find the expression for temperature of processor, $T(t)$ given that $T(0) = 85^\circ\text{C}$. (ii) How long will it take for the processor's temperature to reach 40°C ? Given that $k = 0.03$, $\log_e 4 = 1.3863$.	2 2

A38.	(i) $\frac{dT}{dt} = -k(T - 25)$ $\Rightarrow \frac{dT}{T - 25} = -k dt$ $\Rightarrow \int \frac{dT}{T - 25} = -k \int dt$ $\Rightarrow \log T - 25 = -kt + C \quad \dots(a)$ When $t = 0, T = 85$ $\Rightarrow \log 60 = C$ Using in equation (a), $\log T - 25 = -kt + \log 60 \quad \dots(b)$ (ii) When $k = 0.03, \log T - 25 = -0.03t + \log 60$ $\Rightarrow \log \left \frac{T - 25}{60} \right = -0.03t$ $\Rightarrow T - 25 = 60.e^{-0.03t}$ When $T = 40, t = t_1$ $\Rightarrow \frac{15}{60} = e^{-0.03t_1}$ $\Rightarrow e^{-0.03t_1} = \frac{1}{4} \Rightarrow -0.03t_1 = -\log 4$ $\Rightarrow t_1 = \frac{\log 4}{0.03} = \frac{1.3863}{0.03} = 46.21\text{m}$	$\frac{1}{2}$ 1 $\frac{1}{2}$ 1 $\frac{1}{2}$ $\frac{1}{2}$
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